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Step 10 - Virtual Timer

Reference Note

This chapter is based on concepts described in the ARM Generic Timer Architecture documentation.

The Virtual Timer works exactly like the Physical Timer, except it uses the **Virtual Counter (`CNTVCT_EL0`)** as its time source instead of the **Physical Counter (`CNTPCT_EL0`)**.

The Virtual Timer allows software to generate timer events based on **virtualized time**, enabling virtualization without requiring additional hardware counters.

Learning Objectives

After completing this chapter, you should understand:

- Purpose of the Virtual Timer
- Relationship between Virtual Counter and Virtual Timer
- Virtual Timer registers
- Virtual Timer expiry logic
- Relative vs Absolute programming
- Virtual Timer interrupt generation
- Difference between Physical and Virtual Timers

1. What is the Virtual Timer?

The Virtual Timer generates timer events using the Virtual Counter.

Counter–Timer Relationship

Counter	Timer
CNTPCT_EL0	Physical Timer
CNTVCT_EL0	Virtual Timer

The Virtual Timer operates on:

Virtual Time

instead of:

Physical Time

2. Virtual Timer Architecture

Component	Purpose
System Counter	Master source of time
Virtual Counter (CNTVCT_EL0)	Virtualized view of time
Virtual Timer	Generates events using virtual time
Interrupt Event	Signals timer expiry

Timer Flow

Step	Action
1	Virtual Counter advances
2	Compare logic checks target value
3	Expiry condition becomes true
4	Virtual Timer expires
5	Interrupt event generated

3. Virtual Timer Registers

CNTV_CVAL_EL0

Stores:

Absolute compare value

Question answered:

At what virtual count should the timer expire?

CNTV_TVAL_EL0

Stores:

Relative interval

Question answered:

How many ticks from now?

CNTV_CTL_EL0

Controls timer operation.

Control Fields

Field	Purpose
ENABLE	Start/Stop timer
IMASK	Mask interrupt output
ISTATUS	Timer expiry indication

4. Virtual Timer Expiry Logic

The timer continuously evaluates:

Current Virtual Count
vs
Target Virtual Count

Expiry condition:

CNTVCT >= CNTV_CVAL

When true:

Timer Expires

5. Absolute Programming Example

Current Virtual Counter:

CNTVCT = 1,000,000

Program:

CNTV_CVAL = 1,024,000

Result:

Timer expires when
CNTVCT reaches 1,024,000

Example Table

Parameter	Value
Current CNTVCT	1,000,000
CNTV_CVAL	1,024,000
Delta	24,000 ticks

6. Relative Programming Example

Current Virtual Counter:

CNTVCT = 1,000,000

Program:

CNTV_TVAL = 24,000

Hardware internally calculates:

CNTV_CVAL
=
1,000,000 + 24,000
=
1,024,000

The resulting timer behavior is identical.

7. Virtual Timer Uses Virtual Time

Suppose:

Parameter	Value
CNTPCT	1,000,000
CNTVOFF_EL2	250,000

Then:

CNTVCT = 750,000

Now program:

CNTV_CVAL = 760,000

The timer fires after:

10,000 ticks

because the comparison is performed using:

CNTVCT

not:

CNTPCT

8. Physical vs Virtual Timer

Feature	Physical Timer	Virtual Timer
Counter Source	CNTPCT_EL0	CNTVCT_EL0
Compare Register	CNTP_CVAL_EL0	CNTV_CVAL_EL0
Relative Register	CNTP_TVAL_EL0	CNTV_TVAL_EL0
Control Register	CNTP_CTL_EL0	CNTV_CTL_EL0
Time View	Physical Time	Virtual Time

9. Interrupt Generation

When expiry occurs:

CNTVCT >= CNTV_CVAL

Timer status changes:

ISTATUS = 1

Timer output becomes active.

Eventually:

Virtual Timer
↓

Virtual Timer Interrupt

Interrupt delivery details will be covered later when we discuss:

- PPIs
- GIC
- Interrupt routing

10. Virtual vs Physical Timer Example

Physical Timer

Item	Value
CNTPCT	1,000,000
CNTP_CVAL	1,010,000

Expires after:

10,000 physical ticks

Virtual Timer

Item	Value
CNTVCT	750,000
CNTV_CVAL	760,000

Expires after:

10,000 virtual ticks

Both timers use the same System Counter underneath.

Only the visible time differs.

Architecture Summary

Component	Purpose
System Counter	Actual source of time
CNTVCT_EL0	Virtualized time
CNTV_CVAL_EL0	Virtual deadline
CNTV_TVAL_EL0	Relative interval
CNTV_CTL_EL0	Timer control
Interrupt Event	Notifies software of expiry

Common Misconceptions

Misconception 1

Virtual Timer requires another hardware counter

False.

The Virtual Timer ultimately derives time from the same System Counter.

Misconception 2

Virtual Timer compares against CNTPCT

False.

It compares against:

CNTVCT

Misconception 3

CNTV_TVAL and CNTV_CVAL are different timers

False.

They are two ways to program the same Virtual Timer.

Key Takeaways

Concept	Key Point
Virtual Timer	Uses virtual time
Time Source	CNTVCT_EL0
Absolute Register	CNTV_CVAL_EL0
Relative Register	CNTV_TVAL_EL0
Control Register	CNTV_CTL_EL0
Expiry Condition	CNTVCT >= CNTV_CVAL
Purpose	Generate events using virtualized time

Knowledge Check

Q1

What counter does the Virtual Timer use?

Q2

What is the difference between CNTV_CVAL_EL0 and CNTV_TVAL_EL0?

Q3

Which control register manages the Virtual Timer?

Q4

What condition causes the Virtual Timer to expire?

Q5

Does the Virtual Timer use a separate hardware clock?

Answer Key

Question	Answer
Q1	CNTVCT_EL0
Q2	CVAL is absolute, TVAL is relative
Q3	CNTV_CTL_EL0
Q4	CNTVCT >= CNTV_CVAL
Q5	No, it ultimately derives time from the System Counter